

Servo Arm Control System

1. Product Overview

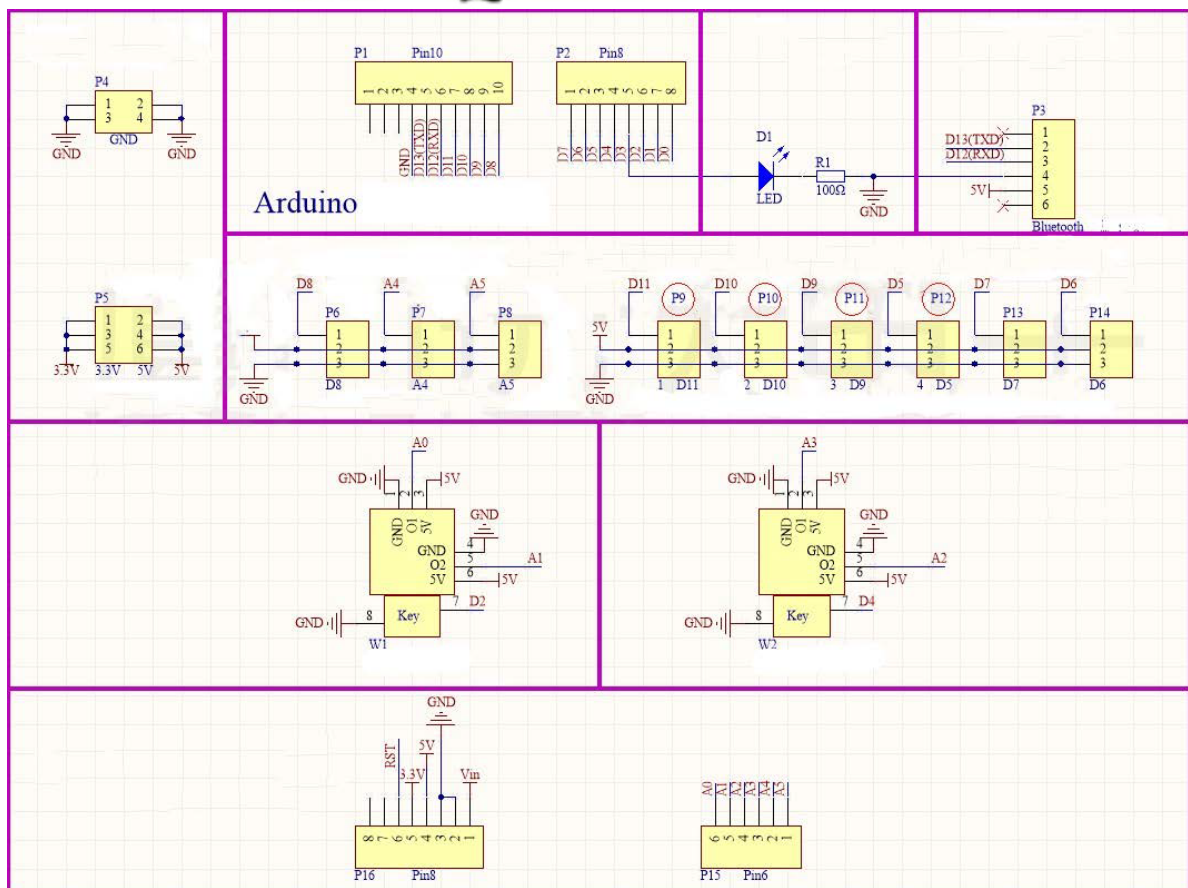
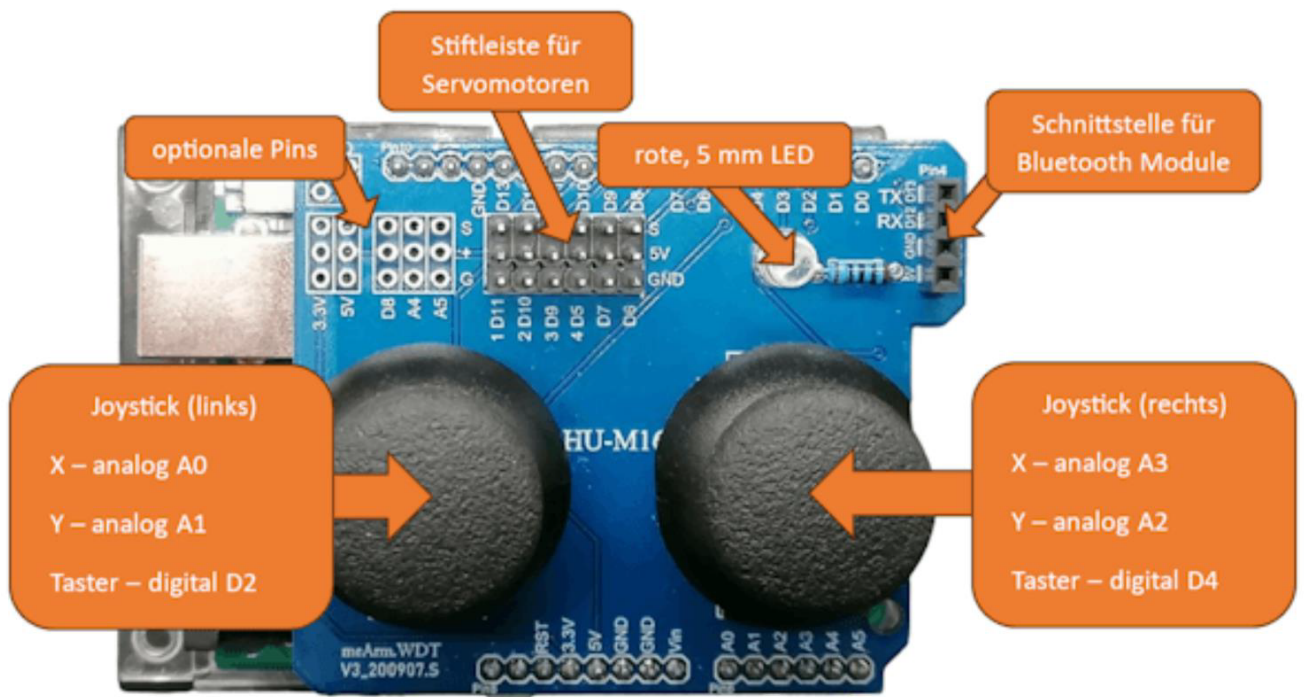
A complete 5-servo robotic arm control system with three control methods:

- Joystick Control - Dual joysticks for real-time control
- Serial Commands - Arduino IDE Serial Monitor
- Python GUI - Computer interface with visual controls

2. Hardware

- Arduino UNO R3 (main controller)
- 2 dual-axis joystick modules
- 5 MG995 servo motors
- Mechanical arm components

3. Pin Definition



4. Control Logic

Control Mapping Table:

Control	Pin	Servo	Range	Function
Joystick 1 X	A0	Servo 0	0-180 deg	X-axis
Joystick 1 Y	A1	Servo 1	0-180 deg	Y-axis
Joystick 1 Btn	D2	Servo 4	0-90 deg	+2 deg/press
Joystick 2 X	A3	Servo 2	0-180 deg	X-axis
Joystick 2 Y	A2	Servo 3	0-180 deg	Y-axis
Joystick 2 Btn	D4	Servo 4	0-90 deg	-2 deg/press

5. Arduino Setup

1. Open servo_control.ino in Arduino IDE
2. Select Board: Arduino UNO
3. Select COM port
4. Upload sketch
5. Open Serial Monitor (9600 baud)

6. Python Environment

Install Python:

Go to <https://www.python.org/downloads/windows/>

Download the Python 3.12.7 Windows installer (64-bit)

Double-click to run the installer. On the first screen, ensure you tick the box:

Add python.exe to PATH (this is very important; if you do not tick this box, the `python` command will not be found later)

Then click 'Customise installation' and ensure that 'tcl/tk' and 'IDLE' are ticked

Install pip and install pyserial:

Press Win + R, type 'cmd', press Enter to open the Command Prompt

Type the following line and press Enter: `pip install pyserial`

Once you see 'Successfully installed pyserial-3.5', the installation is complete

Run: `python servo\_control.py`

7. Power Supply:

After uploading code, switch from computer USB to external power supply (9V 2A or higher). MG995 servos require adequate power.

Overload Protection:

Do not apply excessive force. Servo stall causes current surge and board restart.

Support:

GitHub: <https://github.com/490437477-sys/ARM3>

